

EXPERIMENT 2

Stages of Compilation and Execution of a C Program to Calculate the Area of a Circle

Aim	To observe the compilation stages produced by GCC using -save-temps and execute a C program that calculates the area of a circle.
Formula	$\text{Area} = \pi \times \text{radius} \times \text{radius}$
Software	GCC compiler and a terminal

Procedure

1. Save the program as circle_area.c. 2. Compile it with gcc -save-temps. 3. Observe the temporary files. 4. Run the executable and enter a radius.

C Program

```
#include <stdio.h>

int main(void)
{
    float radius, area;

    printf("Enter the radius: ");
    scanf("%f", &radius);

    area = 3.14159f * radius * radius;
    printf("Area of the circle = %.2f\n", area);

    return 0;
}
```

The program reads the radius, uses $\pi \times r \times r$, and prints the result to two decimal places.

Compilation using -save-temps

Compile the source file with:

```
$ gcc -save-temps -o circle_area circle_area.c
```

The -save-temps option keeps files from the compilation process instead of deleting them.

Files Created

File	Purpose
circle_area.c	Original C source program
circle_area.i	Preprocessed source: header files and macros are expanded
circle_area.s	Assembly-language output
circle_area.o	Object code created from the assembly file
circle_area	Final executable produced after linking

Order of Compilation

```
circle_area.c -> circle_area.i -> circle_area.s -> circle_area.o -> circle_area
```

Preprocessing prepares the source, compilation converts it to assembly, assembling creates object code, and linking produces the executable.

Execution and Result

Run the program from the terminal:

```
$ ./circle_area  
Enter the radius: 5  
Area of the circle = 78.54
```

For radius 5, the calculation is $3.14159 \times 5 \times 5 = 78.54$.

Result

The C program was compiled successfully using `-save-temps`. The intermediate preprocessing, assembly, and object files were retained and the executable correctly calculated the area of a circle.

Conclusion

GCC converts a C program through preprocessing, compilation, assembling, and linking before execution. The `-save-temps` option makes these stages visible for study.